

Supplemental data 1: Compositions of the nutrient solutions derived from Hoagland nutrient solution and modified based on Maillard *et al.* (2016a) in order to get hydroponic ionome as close as field observations (data not shown, see Courbet *et al.* (2021)). These solutions were used to expose 22 days *Brassica napus* L. and *Triticum aestivum* to unique nutrient deprivations in boron (B), chlorine (Cl), manganese (Mn), iron (Fe), nickel (Ni), copper (Cu), zinc (Zn), Molybdenum (Mo), sodium (Na), silicon (Si), cobalt (Co) or selenium (Se).

[illegible]

Nutrients	Elemental concentrations (µM) of deprived solutions :												
	Control	-B	-Cl	-Mn	-Fe	-Ni	-Cu	-Zn	-Mo	-Na	-Si	-Co	-Se
N	3500	3500	3500	3500	3500	3500	3500	3500	3500	3500	3500	3500	3500
Mg	400	400	400	400	400	400	400	400	400	400	400	400	400
P	200	200	500	200	200	200	200	200	200	200	200	200	200
S	507	507	507	504	507	507	506	504	507	507	507	507	507
K	1500	1500	1400	1500	1500	1500	1500	1500	1500	1500	1500	1500	1500
Ca	2110	2110	1610	2110	2110	2110	2110	2110	2110	2110	2110	2110	2110
B	10	0	10	10	10	10	10	10	10	10	10	10	10
Cl	2001	1701	1	1701	1761	1700	1701	1701	1701	1101	1401	1700	1701
Mn	3	3	3	0	3	3	3	3	3	3	3	3	3
Fe	100	100	100	100	0	100	100	100	100	100	100	100	100
Ni	0.15	0.15	0.15	0.15	0.15	0	0.15	0.15	0.15	0.15	0.15	0.15	0.15
Cu	0.7	0.7	0.7	0.7	0.7	0.7	0	0.7	0.7	0.7	0.7	0.7	0.7